CRIPPLED SPACESHIP

Solo Survival Board Game — Design Specification

Working titles: LAST VECTOR • DEAD RECKONING • PERIAPSIS • THE LONG FALL

A deep-space vessel suffers a catastrophic accident far from home. The crew cannot restore it completely. Survival requires repeatedly choosing which systems to preserve, overload, damage, or cannibalise to solve the crisis immediately ahead.

"Every problem is solvable. You just might need the solution later."

1. Design Goals

The game preserves the downward trajectory of Resist!, but makes that mechanism native to spacecraft survival. The ship itself is the deteriorating deck.

- Solo-first design; target play time 30–45 minutes.
- Powerful actions solve immediate problems by reducing future capability.
- Failures create persistent states and cascading failures rather than merely subtracting points.
- Most crises should be solvable, but solving them may create strategically disastrous future weaknesses.
- The table should visibly tell the story of a spacecraft being dismantled to survive.

2. Narrative Arc

CAPABLE SHIP → DAMAGED SHIP → IMPROVISED SHIP → LIFEBOAT → FINAL RETURN

The objective is to return to safety with as many crew alive as possible and, secondarily, with as much of the spacecraft, mission data, and remaining resources intact as possible.

3. Core Mechanic: Normal, Redline, Sacrifice

System cards replace Resist!'s characters. Each can be used at one of three commitment levels:

- NORMAL — modest, repeatable effect; the system remains operational.
- REDLINE — powerful effect; flip the system to its damaged side.
- SACRIFICE — exceptional effect; permanently destroy or remove the system.

MAIN REACTOR
NORMAL — Power 3; Heat +1.
REDLINE — Power 7; Heat +3; damage this system.
SACRIFICE — Power 10; permanently destroy the reactor.

The central question is not simply whether the current crisis can be solved, but which future capability the player is willing to lose in order to solve it.

4. The Ship as the Deck

The spacecraft begins with roughly 18 system cards. Only a subset—perhaps five—is available during each crisis, representing routing, access, power availability and emergency chaos.

System	Normal	Emergency / Redline
Main Reactor	Power 3	Power 7
Auxiliary Reactor	Power 2	Power 4
Main Engine	Thrust 3	Thrust 7
Manoeuvring Thrusters	Thrust 1 / Control 2	Control 5
Navigation Computer	Navigation 3	Navigation 6
Main Computer	Compute 3 / Control 1	Compute 6
Communications	Signal 3	Signal 6
Life Support	Life Support 3	Life Support 6
Thermal Control	Cooling 3	Cooling 6
Sensors	Detection 3	Detection 6
Medical Bay	Medical 3	Medical 6
Fabricator	Engineering 2	Engineering 5
Batteries	Power 2	Power 5
Fuel Cells	Power 2	Power 4
Radiation Shielding	Protection 3	Protection 6
Cargo Bay	Storage	Sacrifice cargo
Habitat	Crew capacity	Emergency shelter
Service Module	Engineering 2	Cannibalise

5. Capabilities

Crises should demand combinations of capabilities rather than a single resource. Initial candidates are:

- POWER
- ENGINEERING
- COMPUTE
- THRUST
- LIFE SUPPORT
- CONTROL
- MEDICAL (optional seventh capability)

Systems can provide multiple capabilities, encouraging combinations and improvisation rather than simple symbol matching.

6. Crisis Cards

Crisis cards replace missions. Each specifies requirements and a persistent consequence for failure. Successful resolutions may sometimes provide a repair, resource or positional benefit.

COOLANT LOOP RUPTURE
Requirement: Engineering 5 + Power 3.
Failure: Heat +2.
Success: patch one damaged Thermal system.

CO_2 BUILDUP
Requirement: Life Support 6.
Failure: injure one crew member and create an ongoing atmospheric penalty.

NAVIGATION DRIFT
Requirement: Compute 4 + Control 4.
Failure: add one Course Error.

MICROMETEOROID IMPACT
Requirement: Engineering 5 + Control 2.
Failure: damage two random systems.

7. Cascading Failure

Failure should usually create a new state rather than simply deduct points. Problems interact and propagate.

Cooling failure → Heat rises → Reactor scram → Power shortage → Life support failure → Crew injury

A compromise several turns earlier can therefore become the origin of a later catastrophe. The player should be able to trace the history of the disaster through the current ship state.

8. Global Tracks

- HEAT: 0 → 1 → 2 → 3 → 4 → 5 → CRITICAL.
- OXYGEN: 6 → 5 → 4 → 3 → 2 → 1 → DEAD.
- COURSE ERROR: 0 → 1 → 2 → 3 → LOST.
- DISTANCE HOME: e.g. 10 → 0; the principal game clock.
- POWER should preferably be generated and spent within a crisis rather than maintained as another persistent track.

9. Crew

Crew matter emotionally and tactically without replacing the spacecraft as the main engine. A six-person tableau might contain Commander, Pilot, Engineer, Systems Engineer, Doctor and Mission Specialist.

Crew states are Healthy → Injured → Dead. Each character has a modest once-per-crisis or situational ability; injury weakens or removes it.

ENGINEER
Once per crisis: +2 Engineering.
Possible specialist variant: gain one extra component whenever a system is cannibalised.

10. Cannibalisation

Cannibalisation is a signature mechanism. A system can be deliberately destroyed to obtain components, temporary capability or repairs elsewhere.

CANNIBALISE COMMUNICATIONS
Permanently destroy Communications. Gain 2 Electronics, 1 Wiring and 1 Power Cell. Use these components to patch or repair other systems.

The payoff comes later when a crisis demands a system the player previously judged expendable. Choices should create delayed consequences rather than immediately announcing themselves as correct or incorrect.

11. Journey Structure and Hidden Information

Divide the journey into three escalating crisis decks:

- PHASE I — ACCIDENT: fire, decompression, reactor trip, coolant leak, electrical short.
- PHASE II — TRANSIT: radiation storm, course correction, CO_2 accumulation, battery degradation, micrometeoroids, illness.
- PHASE III — RETURN: communications acquisition, precision navigation, re-entry control, heat shielding, landing or recovery.

Use only a subset of each deck per game. Players learn the kinds of threats associated with each phase without knowing exactly which crises will occur.

12. Partial Forecasting

Reveal only the category of the next crisis—ENVIRONMENT, NAVIGATION, MEDICAL, ENGINEERING or ELECTRICAL—while keeping the exact card hidden. Sacrifices therefore become informed gambles rather than blind guesses.

13. Route Decisions: Push Your Luck

Replace Resist!'s stop-or-continue decision with trajectory choices. Short routes reduce oxygen consumption and crisis exposure but demand precious capability immediately.

RETURN TRAJECTORY
SAFE TRAJECTORY — add 3 journey cards; low immediate thrust requirement. DIRECT RETURN — add only 1 journey card; immediately meet Thrust 8.

14. Damage and Repair

- PATCH — temporary or limited restoration.
- REPAIR — restore a damaged system using scarce spare parts.

- CANNIBALISE — destroy one system to repair or support another.

System cards should ideally be double-sided: OPERATIONAL → DAMAGED. A further severe use or hit destroys the system. Begin with only a small pool of true spare parts, perhaps six.

15. Visual Deterioration

The physical table state should visibly deteriorate:

- Start: approximately 18 operational systems.
- Mid-game example: 11 operational, 5 damaged, 2 destroyed.
- Late-game example: 5 operational, 4 damaged, 9 destroyed.

The tableau itself should become a visual history of the emergency.

16. Spacecraft Variants

Ship	Type	Character
ODYSSEY	Exploration vessel	Balanced; 18 systems, 6 crew, 6 spares.
HERMES	Scientific vessel	Excellent computing and sensors; poor redundancy.
HERCULES	Cargo vessel	Large stores; poor manoeuvrability.
ICARUS	Experimental nuclear vessel	Enormous power; severe heat problems.

17. Turn Structure

1. CONSUME — reduce oxygen / advance time as required.
2. CRISIS — reveal the current Crisis card.
3. ASSEMBLE — draw approximately five available System cards.
4. RESPOND — use systems in Normal, Redline or Sacrifice mode and apply crew abilities.
5. RESOLVE — meet requirements or suffer the crisis consequence.
6. DAMAGE — apply heat, injuries, system damage and persistent effects.
7. NAVIGATE — advance toward home if trajectory permits.

18. Worked Example

The ship is six days from Earth. Oxygen is 4, Heat is 3 and Course Error is 1. Main Reactor is damaged, Communications are destroyed, Navigation and Batteries are operational, and Life Support is damaged.

OXYGEN SCRUBBER FAILURE
Requirement: Life Support 5 + Engineering 2.
Failure: Oxygen −2 on every subsequent turn until repaired.

The available systems provide Life Support 3 and Engineering 2. The player is two Life Support short. They can Redline Life Support and destroy it after the crisis, or cannibalise the Navigation Computer for parts and repair Life Support. Navigation may, however, be essential during re-entry.

The decision is not "Can I beat this card?" but "Which future disaster am I willing to create to survive this one?"

19. Endgame and Scoring

Reaching Earth is the primary success condition, but survival should have degrees.

- +10 — Reach Earth.
- +3 — Each surviving crew member.
- +1 — Each functioning system.
- +1 — Each unused spare.
- +5 — Recover scientific / mission data.
- +5 — Recover the spacecraft rather than abandoning it.
- Optional penalties for injuries, destroyed critical systems or lost mission objectives.

Example outcomes: ODYSSEY RETURNED (all crew alive, spacecraft recovered); EMERGENCY RE-ENTRY (partial crew survival, command section recovered); CONTACT LOST (trajectory missed or crew lost).

20. Prototype Component List

- 18–24 double-sided System cards.
- 6 Crew cards.
- 30–45 Crisis cards divided across three phases.
- 8–12 Trajectory / Decision cards.
- 1 spacecraft status board.
- Heat, Oxygen, Course Error and Distance markers.
- A small pool of Spare Part / Component tokens.
- Damage / patch markers if double-sided cards alone are insufficient.

21. Design Principles for Playtesting

- A crisis should often be technically solvable; the interesting question is its cost.
- No single system should be obviously expendable in every game.
- Late crises should make early sacrifices echo without making outcomes fully predictable.
- Repair must remain scarce enough that deterioration is fundamentally irreversible.
- Bookkeeping should stay light; complexity should emerge from interactions among systems and persistent failures.
- The strongest decisions should trade immediate certainty against future flexibility.
- The player should finish successful games with a ship that looks and feels barely capable of completing the journey.

22. Core Design Statement

You do not build an engine. You dismantle one.

The game is about controlled deterioration. Each emergency asks the player to convert long-term capability into short-term survival. The accumulating damage, missing systems, injured crew and narrowing options create the narrative. The ideal final turn is not a pristine spacecraft completing a routine landing, but a machine held together by patches and irreversible compromises making one last manoeuvre home.